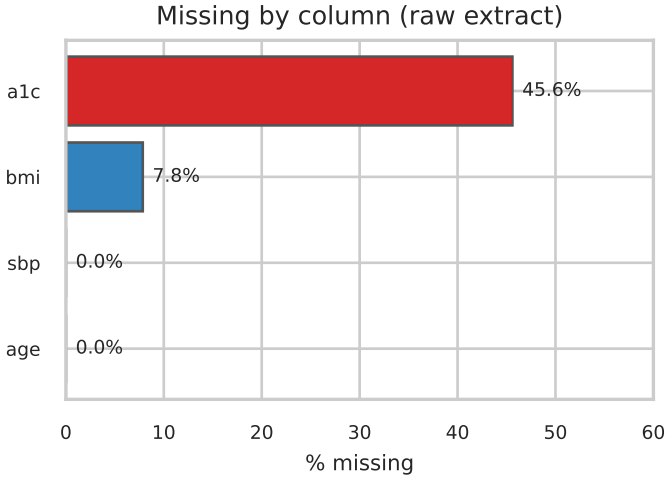
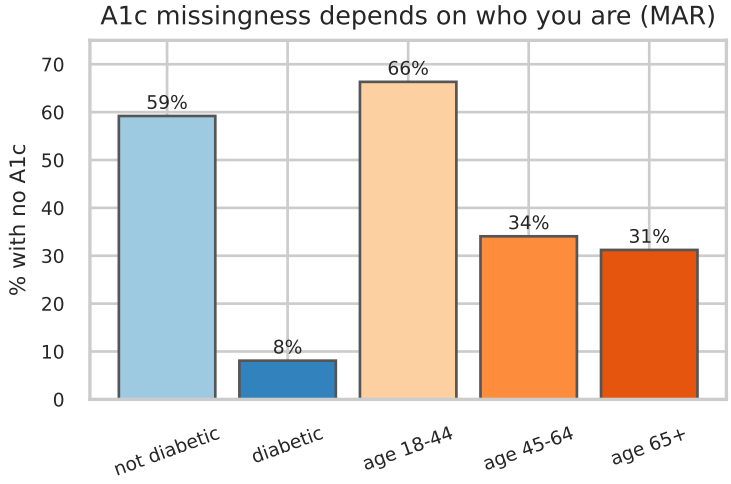
Day 11 Cheat Sheet: Missing Data and Outliers

Patient labs and vitals: find the holes, the impossible values, and fix them honestly

TERM	WHAT IT MEANS	CODE
Missing rate	share of NaN per column	`df.isna().mean()`
MCAR	missing purely at random (scale skipped)	missing rate flat across groups
MAR	missing depends on OTHER columns (age, diabetic)	`df.groupby(g)[col].apply(lambda s: s.isna().mean())`
MNAR	missing depends on the value itself	can't test from data; ask how it's collected
Complete case	drop rows with NaN (pandas does it silently)	`df[col].mean()` skips NaN
Mean imputation	fill with one number: same mean, smaller spread	`s.fillna(s.mean())`
Group imputation	fill within similar patients	`s.fillna(df.groupby(g)[col].transform("mean"))`
Sentinel value	a fake code for "not taken" (0, 999, -1)	`s.replace([0, 999], np.nan)`
IQR rule	outside Q1-1.5*IQR .. Q3+1.5*IQR	flags tails, even real ones
Robust z (MAD)	median-based z, not dragged by the outliers	`0.6745*(x-med)/MAD > 3.5`
Domain rule	clinically impossible range	BMI 12-80, SBP 60-260
Missing flag	keep "was missing" as a model feature	`SimpleImputer(add_indicator=True)`





THIS DATA (simulated, seed 11): 2,000 patients; A1c missing 45.6%, BMI 7.8%
A1c missing for 59% of non-diabetics vs 8% of diabetics; 66% of 18-44 vs 31% of 65+
Impossible values: 6 BMIs over 200 (dropped decimal), 6 SBPs coded 0 or 999 (sentinels)
Mean A1c: truth 6.06, complete case 6.45, group-imputed 6.06
Share with A1c >= 8: truth 11.1%, among tested 18.8% (about 1.7 times)

HABITS THAT KEEP YOU HONEST
- Run isna().mean() and describe() before any chart: max BMI 427 is a typo, not a patient
- Ask WHY it's missing: compare patients with and without the value
- Fix what you can explain (a dropped decimal); set what you can't to NaN
- Never delete outliers just because a rule flagged them: check them against clinical ranges
- In ML, fit the imputer on training rows only and keep a missing-flag column

```
df = pd.read_csv("patients_labs.csv"); print(df.isna().mean(), df.describe())
df["sbp"] = df["sbp"].replace([0, 999], np.nan)          # sentinels -> missing
df.loc[df.bmi > 80, "bmi"] /= 10                         # dropped decimal point
g = df.groupby(["diabetic", "age_band"])["a1c"].transform("mean"); df["a1c_imp"] = df["a1c"].fillna(g)
```

Infographic: Holes and Typos in Patient Data

Simulated: 2,000 primary care patients, A1c, BMI and blood pressure

Profile	Why?	Rules	Fix	Impute	Flag
isna(), describe()	compare with vs without	clinical ranges	explain it or NaN it	group / KNN, not one value	keep a missing flag

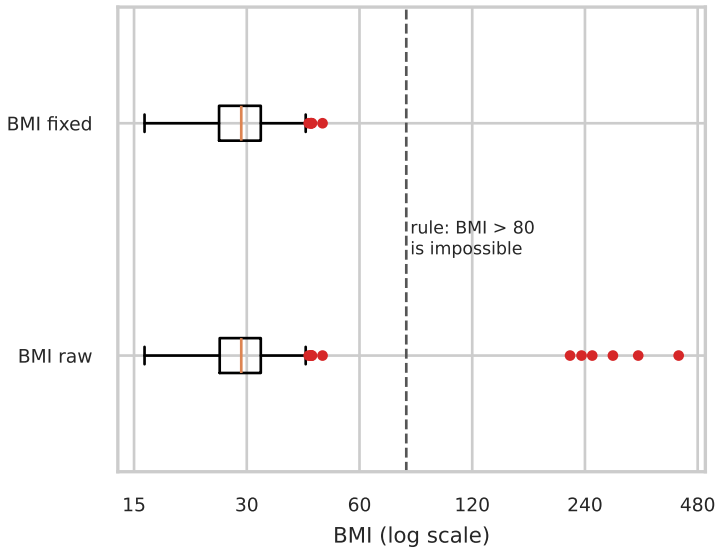
A1C MISSING
46%
of patients

TESTED A1c
6.45 vs 6.06
complete case vs truth

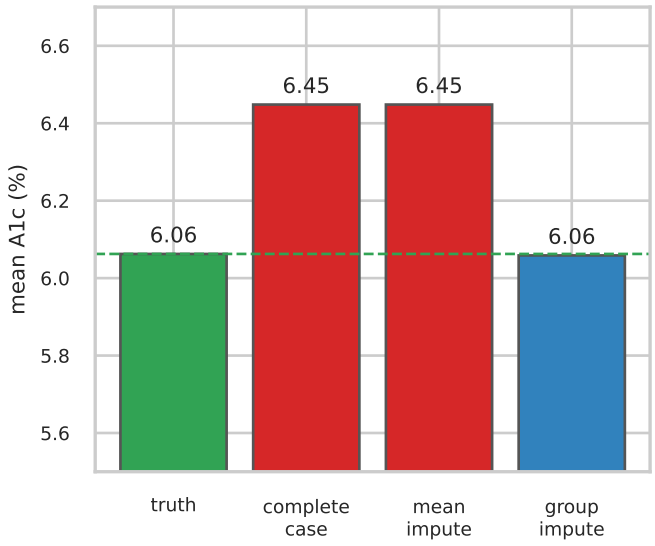
IMPOSSIBLE
12 values
BMI and SBP

KNN vs MEDIAN
0.39 vs 0.58
imputation error

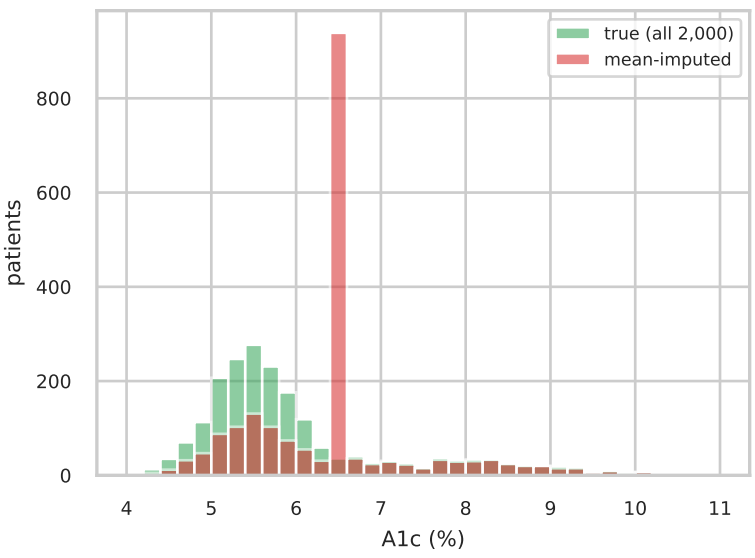
6 dropped decimals: 31.2 typed as 312



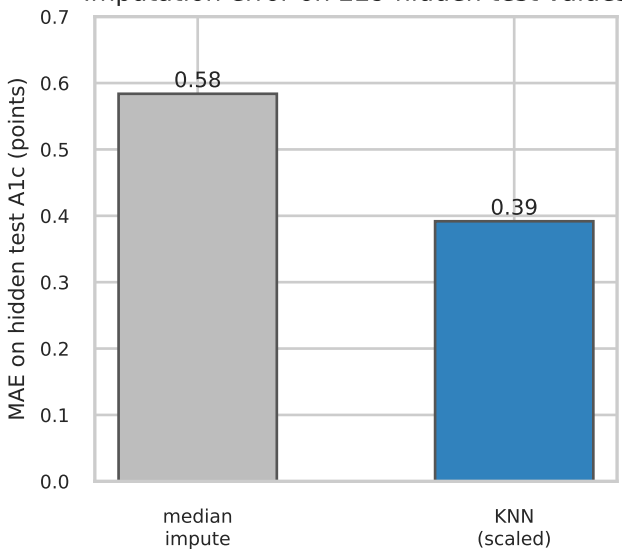
Mean A1c: who got tested biases the answer



Mean imputation: a fake spike, shrunk spread



Imputation error on 229 hidden test values



Read it like a quality manager: doctors order A1c mostly for patients they already worry about, so the tested group looks sicker. Tested-only 'A1c >= 8' runs about 1.7 times the true rate. Impute within similar patients and say what you did. Simulated data.

Hands-On Worksheet: ds-genai-90-days / day11

60-90 minutes | notebook: day11/missing_outliers.ipynb

[ ] mkdir day11, conda activate ds90, open missing_outliers.ipynb

[ ] Simulate patients_labs.csv with seed 11 (snippet 1); keep 'truth' aside

[ ] isna().mean() and describe(): list every red flag you see

[ ] A1c missing rate by diabetic status and age band (snippet 2)

[ ] Compare age, SBP and % diabetic for tested vs not tested

[ ] Check BMI missingness is flat across age bands (MCAR)

[ ] Flag outliers 3 ways: IQR, z > 3, robust z (snippet 3)

[ ] Apply domain rules: BMI 12-80, SBP 60-260; list the bad values

[ ] Fix dropped decimals; set sentinels 0 / 999 to NaN (snippet 4)

[ ] Mean A1c: truth vs complete case vs mean vs group impute

[ ] Plot true vs mean-imputed A1c histograms; note the spike

[ ] Train/test split; SimpleImputer(add_indicator) + scaled KNN (snippet 5)

[ ] Findings cell; git add day11 && git commit && git push

[ ] Stretch: Snowflake data-quality query (null rates, rule breaks)

```
import numpy as np, pandas as pd
df = pd.read_csv("patients_labs.csv")
print(df.isna().mean().round(3))                          # A1c ~0.46
print(df.groupby("diabetic")["a1c"].apply(lambda s: s.isna().mean()))
med = df.bmi.median(); mad = (df.bmi - med).abs().median()
print(((0.6745 * (df.bmi - med) / mad).abs() > 3.5).sum())  # robust z flags
df["sbp"] = df["sbp"].replace([0, 999], np.nan)
```

A1c missing %: _____ Non-diabetic missing %: _____ Diabetic missing %: _____ Impossible BMIs: _____

True mean A1c: _____ Complete case: _____ Group impute: _____ MAE median: _____ MAE KNN: _____

Day 11 Self-Check Quiz

1. Why run isna().mean() and describe() before making any chart?

_____

2. BMI is missing about 8% in every age band. Which missingness type is that?

_____

3. A1c is missing for 59% of non-diabetics but 8% of diabetics. Why does it matter?

_____

4. Mean imputation keeps the mean of the tested patients. What goes wrong?

_____

5. An SBP of 999 shows up 3 times. Outlier to delete, or something else?

_____

6. The IQR rule flags 20 SBPs but only 6 are impossible. What about the other 14?

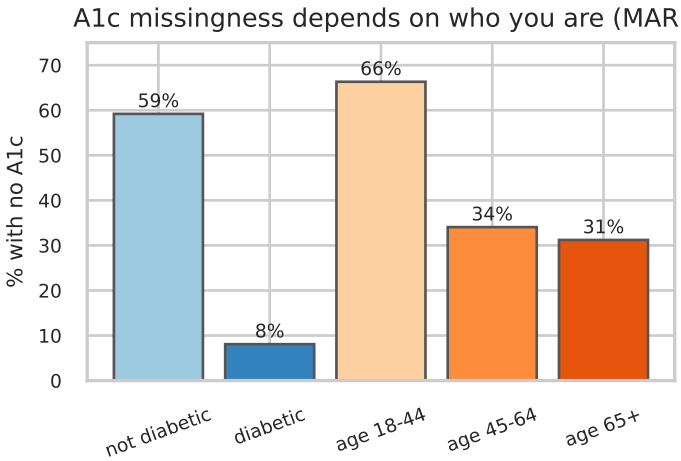
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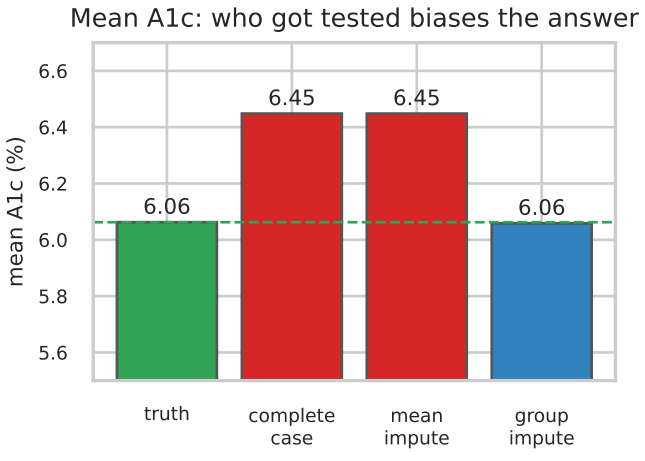
7. Why fit the imputer on training rows only?

_____

8. Why scale features before KNN imputation?

_____





Answers:
1 they show holes and impossible values (BMI 427, SBP 999) in seconds; a chart drawn on dirty data hides or exaggerates them
2 MCAR (missing completely at random): the rate doesn't depend on any column, so complete-case results stay unbiased, just less precise
3 it's MAR: tested patients are older and far more often diabetic, so tested-only stats look sicker (mean 6.45 vs true 6.06)
4 it repeats the biased tested mean for every untested patient, adds a fake spike and shrinks the spread (SD 1.20 to 1.03)
5 a sentinel code for "not taken": replace it with NaN, then treat it as missing; it is not a real blood pressure
6 they are real but unusual readings (very high or low BP). Keep them; check clinically, don't delete just because a rule flagged them
7 test rows must stay unseen; imputing with all rows leaks test information into training and makes the model look better than it is
8 KNN uses distances; unscaled, age (18-89) and SBP (80-200) swamp A1c and diabetic status, so neighbours are chosen badly